

Object-oriented Design with C#

Introduction to Object-oriented Design



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Version Check



This course was created by using:

- C# 10
- Visual Studio 2022 (v17.4.0)
- Markdown Editor v2 (v2.0.130)
- NUnit 3 (v3.13.3)
 - NUnit3TestAdapter (v4.3.0)



Version Check

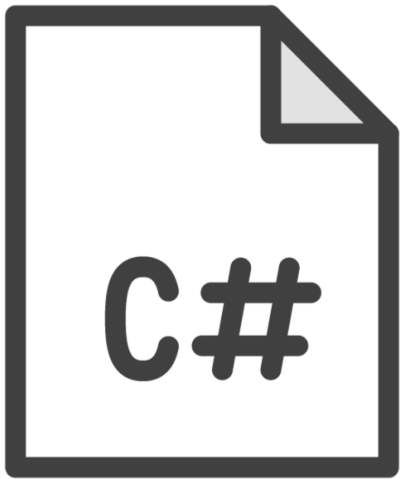


This course is 100% applicable to:

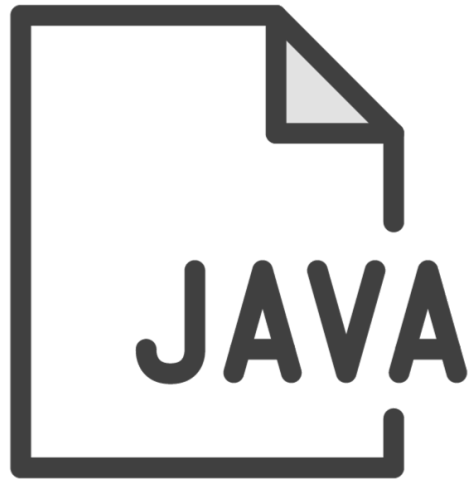
- C# 12
- Visual Studio 2022 (v17.8.4)
- Markdown Editor v2 (v2.0.149)
- NUnit (v3.14.0)
 - NUnit3TestAdapter (v4.5.0)



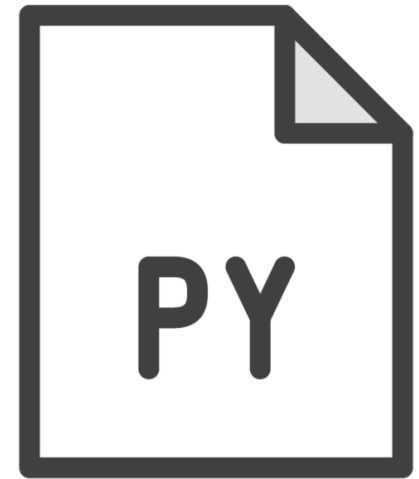
Object-oriented Language Examples



C#



Java



Python



Ruby



PHP



Visual Basic



Object-oriented Programming

!= Object-oriented Design

!= Object-oriented Language



Procedural C#

```
Main()
```

```
{
```

```
    _____
```

```
    _____
```

```
        _____
```

```
        _____
```

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        _____
```

```
    _____
```

```
    _____
```

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        _____
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        _____
```

```
        _____
```

```
    _____
```

```
}
```

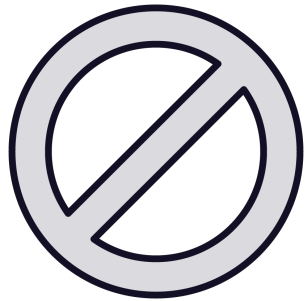


Object-oriented Basic

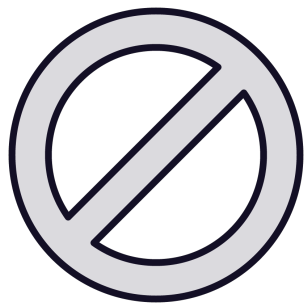
Program



Object-oriented Languages



Best fit for implementing object-oriented designs



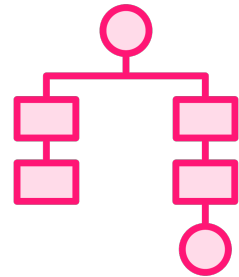
Can't do it all for you



Object-oriented Design



Thinking in objects



Commitment to organization of responsibilities



Classes are tools, used by other classes with other responsibilities



Demo

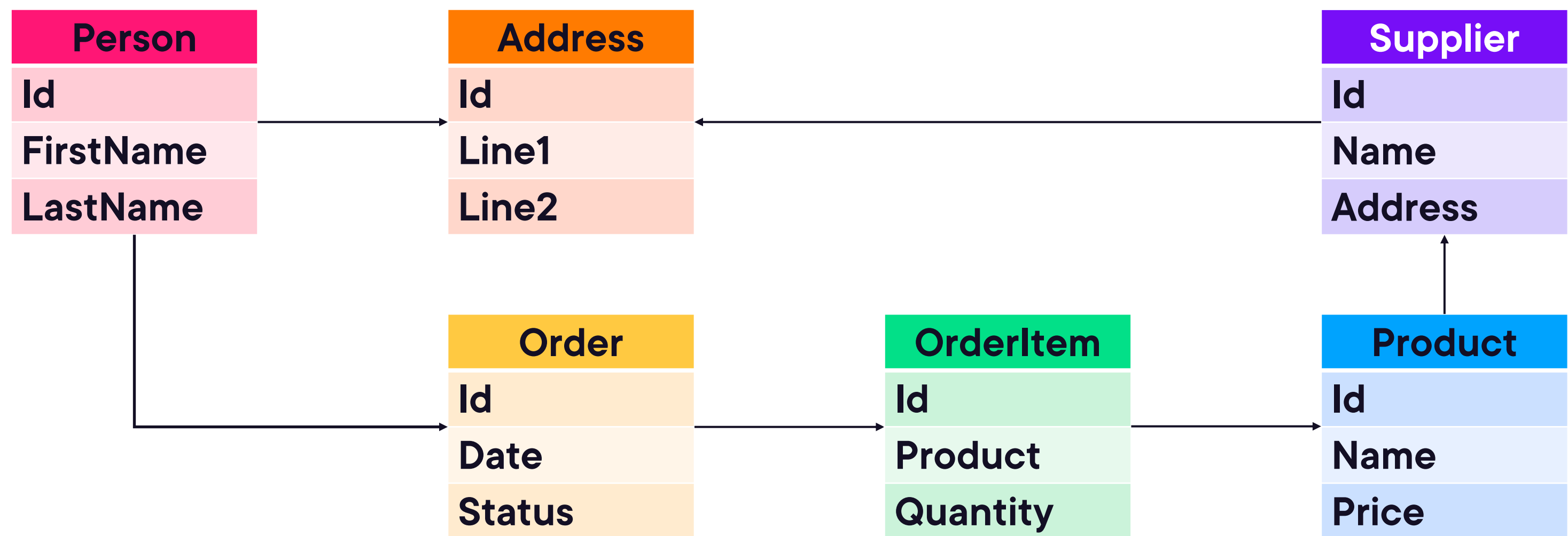


Mermaid

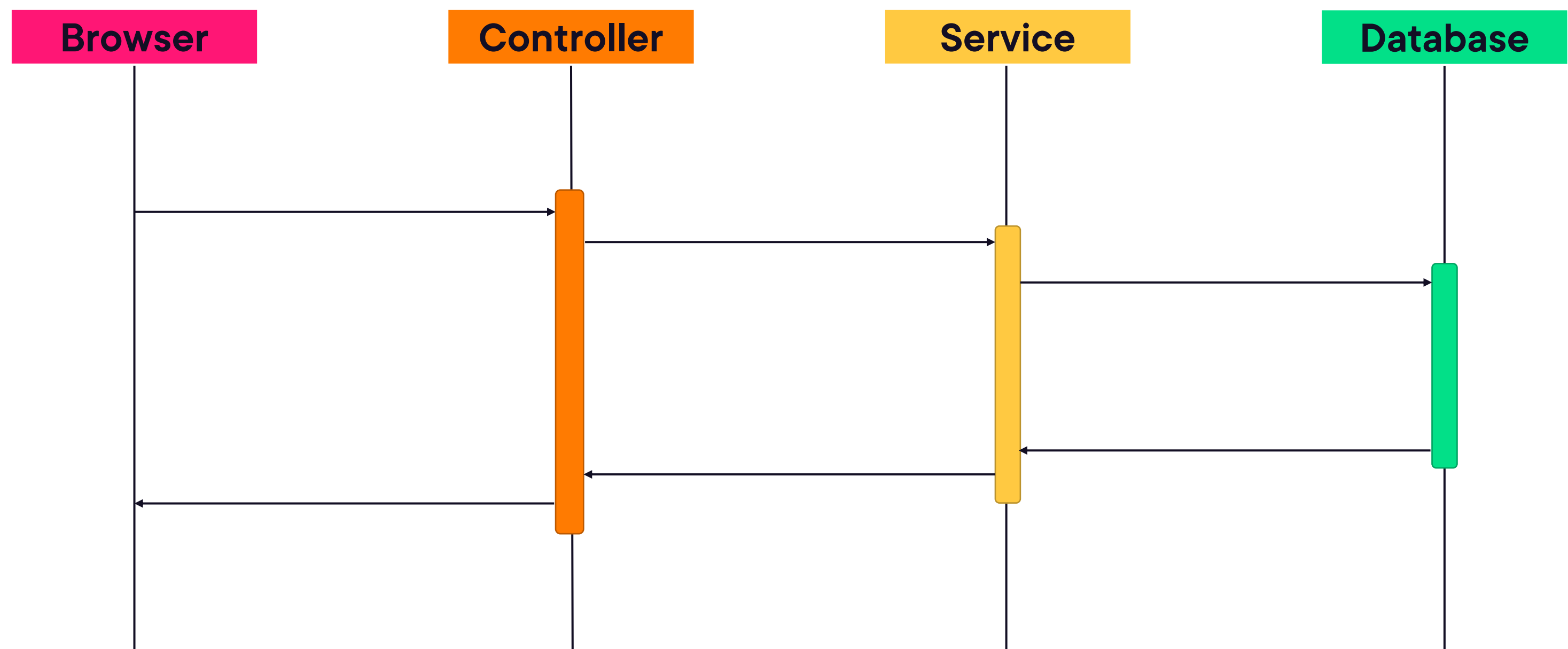
- UML diagrams right in your documentation
- Editable from within your IDE



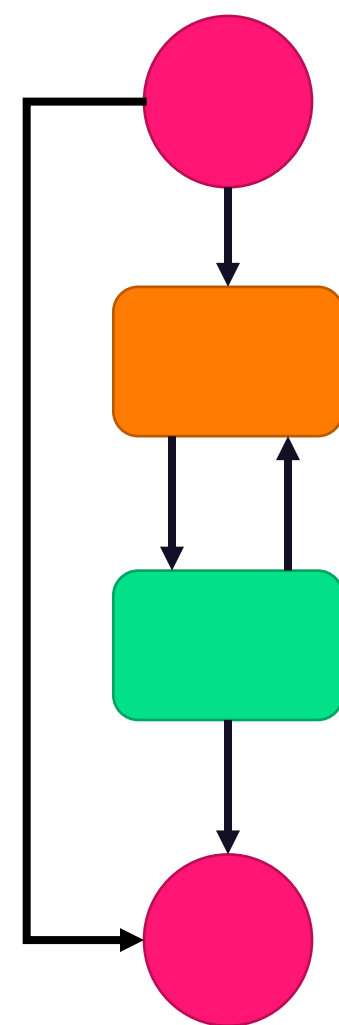
Class Diagram



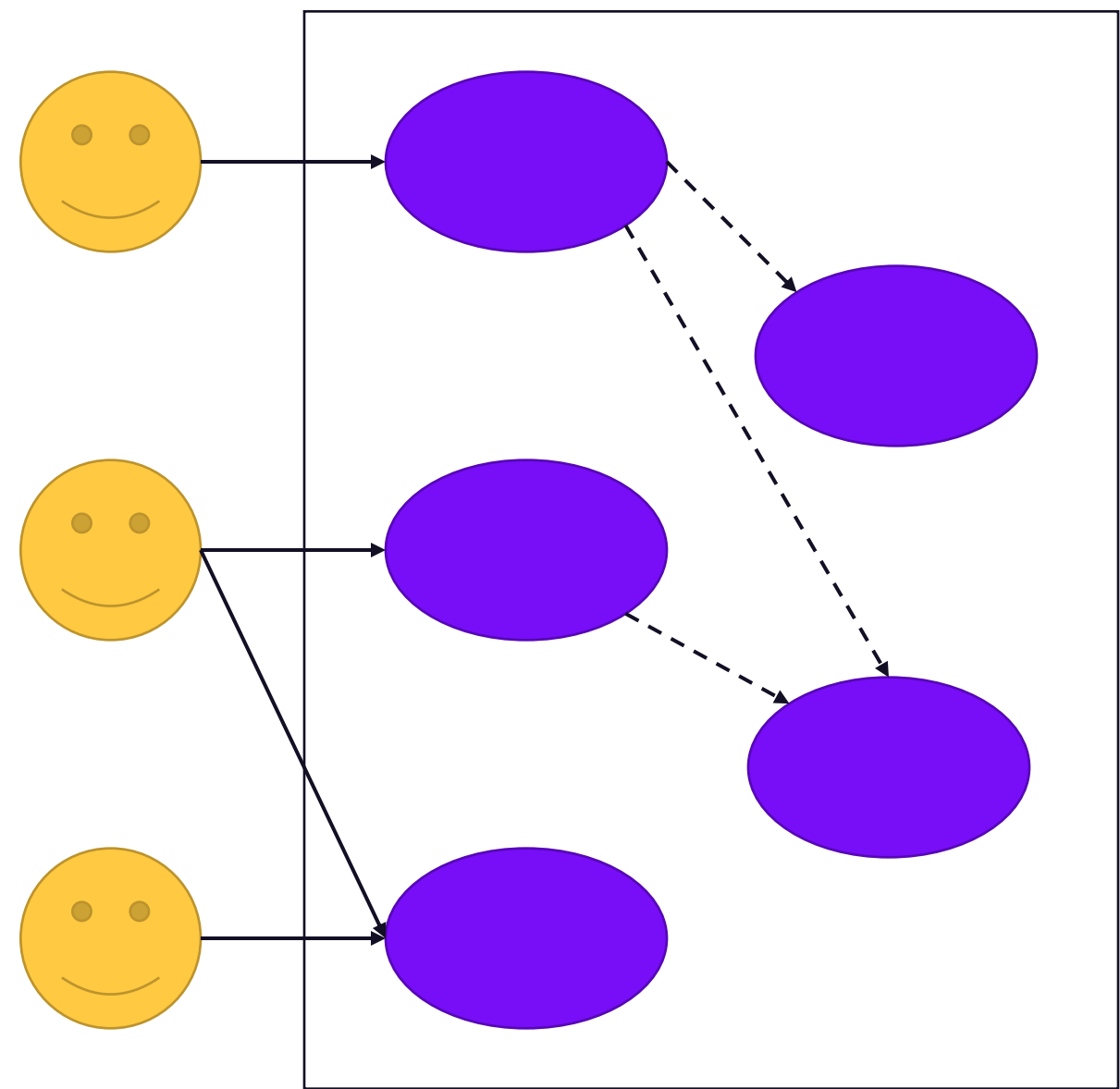
Sequence Diagram



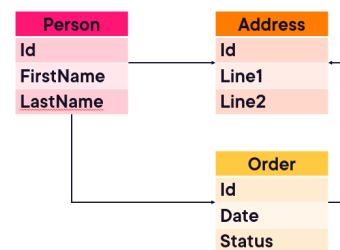
State Diagram



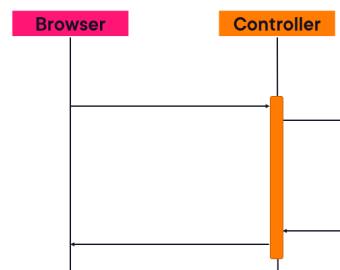
Use Case Diagram



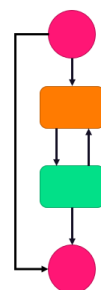
Unified Modeling Language (UML)



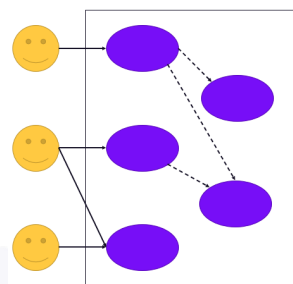
Class diagrams



Sequence diagrams



State diagram



Use case diagram



Mermaid-js

Type to search

mermaid

Introduction

About Mermaid

Deployment

Syntax and Configuration

Diagram Syntax

Flowchart

Sequence Diagram

Class Diagram

State Diagram

Entity Relationship Diagram

User Journey

Gantt

Pie Chart

Requirement Diagram

Gitgraph (Git) Diagram

C4C Diagram (Context) Diagram

Other Examples

Deployment and Configuration

Tutorials

API-Usage

Mermaid API Configuration

Directives

Theming

[Edit this Page](#)

About Mermaid

Mermaid lets you create diagrams and visualizations using text and code.

It is a JavaScript based diagramming and charting tool that renders Markdown-inspired text definitions to create and modify diagrams dynamically.

If you are familiar with Markdown you should have no problem learning [Mermaid's Syntax](#).

The collage features several diagrams: a Gantt chart titled 'Adding GANTT diagram functionality to mermaid' showing task bars for 'Completed task', 'Active task', and 'Future task'; a flowchart with nodes 'Square Rect', 'Circle', 'Rhombus', and 'Round Rect' connected by arrows; a sequence diagram with participants 'Alice', 'Bob', and 'John the Long' showing a conversation with loops; and a state diagram with states 'A section', 'Critical tasks', 'Documentation', and 'Last section'.

build passing

npm v9.1.7

coverage 45%

slack

join chat



Object-oriented Design Is



~~Code written in an object-oriented language~~



~~Code that uses language features like inheritance and interfaces~~



Planning out a system in terms of objects and their responsibilities

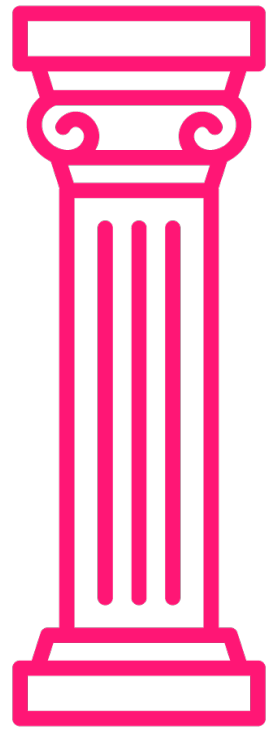




Pillars and Principles

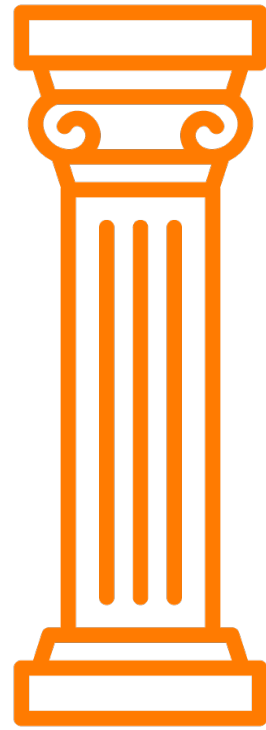


The Four Pillars



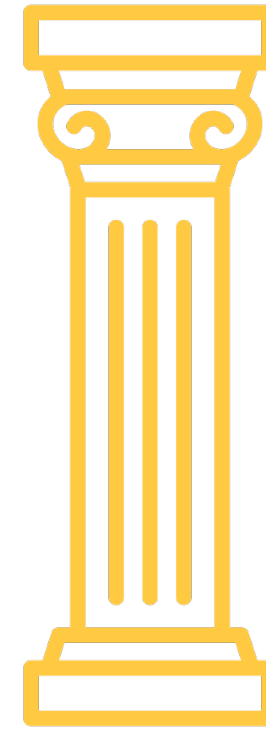
Abstraction

**Implementation
Hiding**



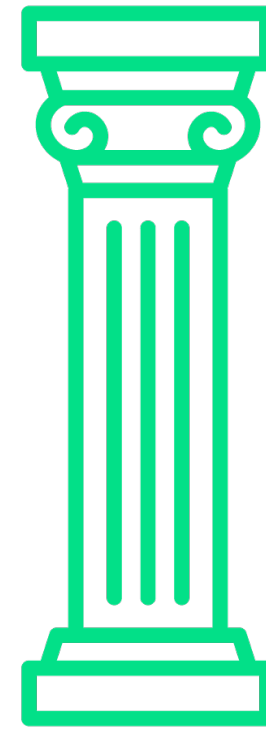
Encapsulation

**Data
Hiding**



Inheritance

**Object
Taxonomy**



Polymorphism

**Object
Interchangeability**



The SOLID Principles

S

Single Responsibility Principle

O

Open-Closed Principle

L

Liskov Substitution Principle

I

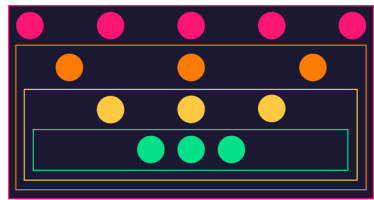
Interface Segregation Principle

D

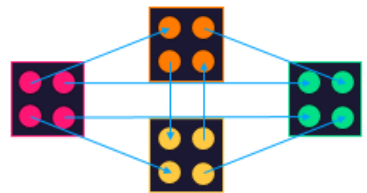
Dependency Inversion Principle



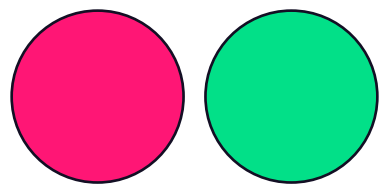
Other Object-oriented Vocabulary



Cohesion



Coupling



Orthogonality

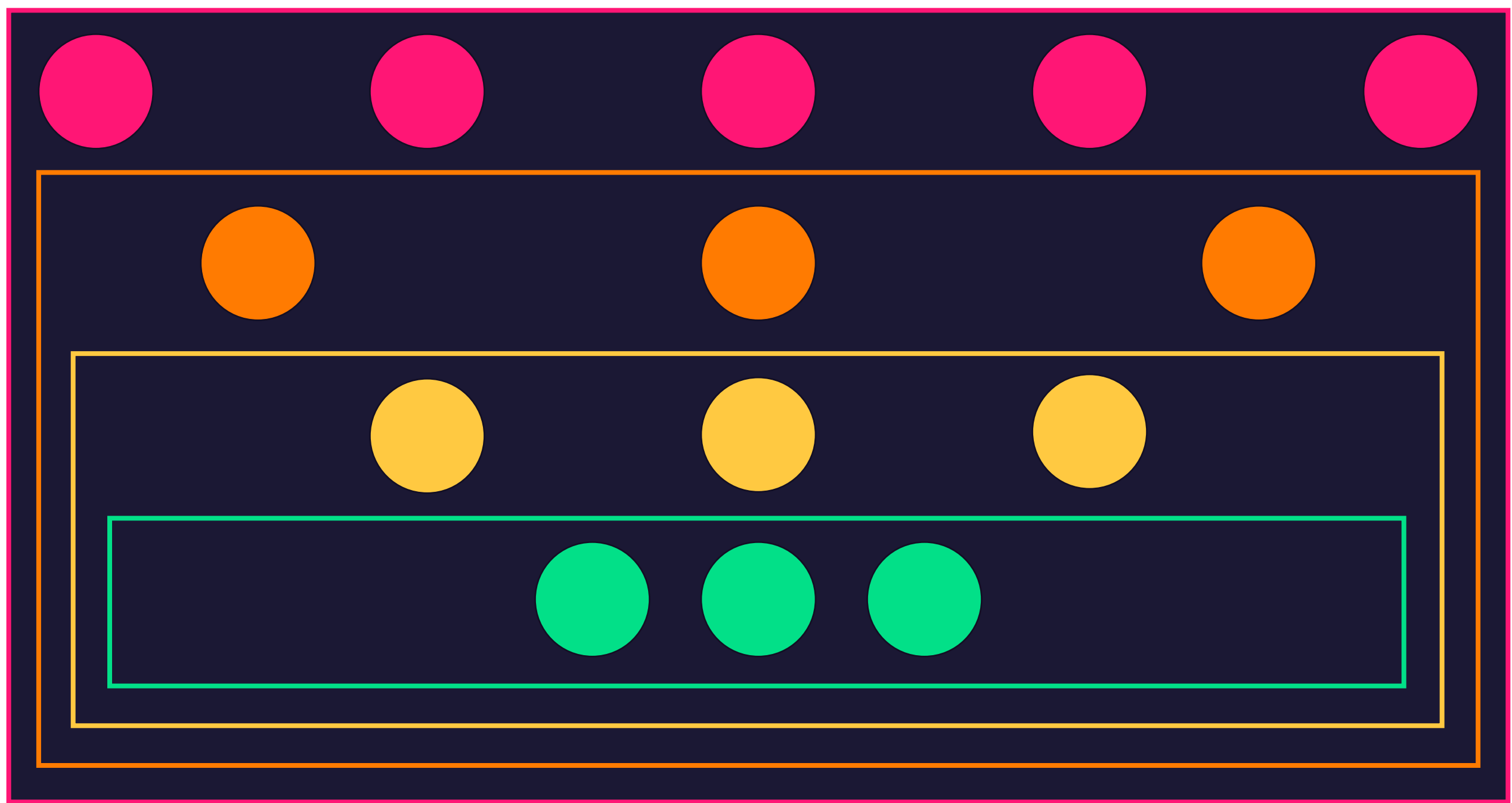




Cohesion



Cohesion

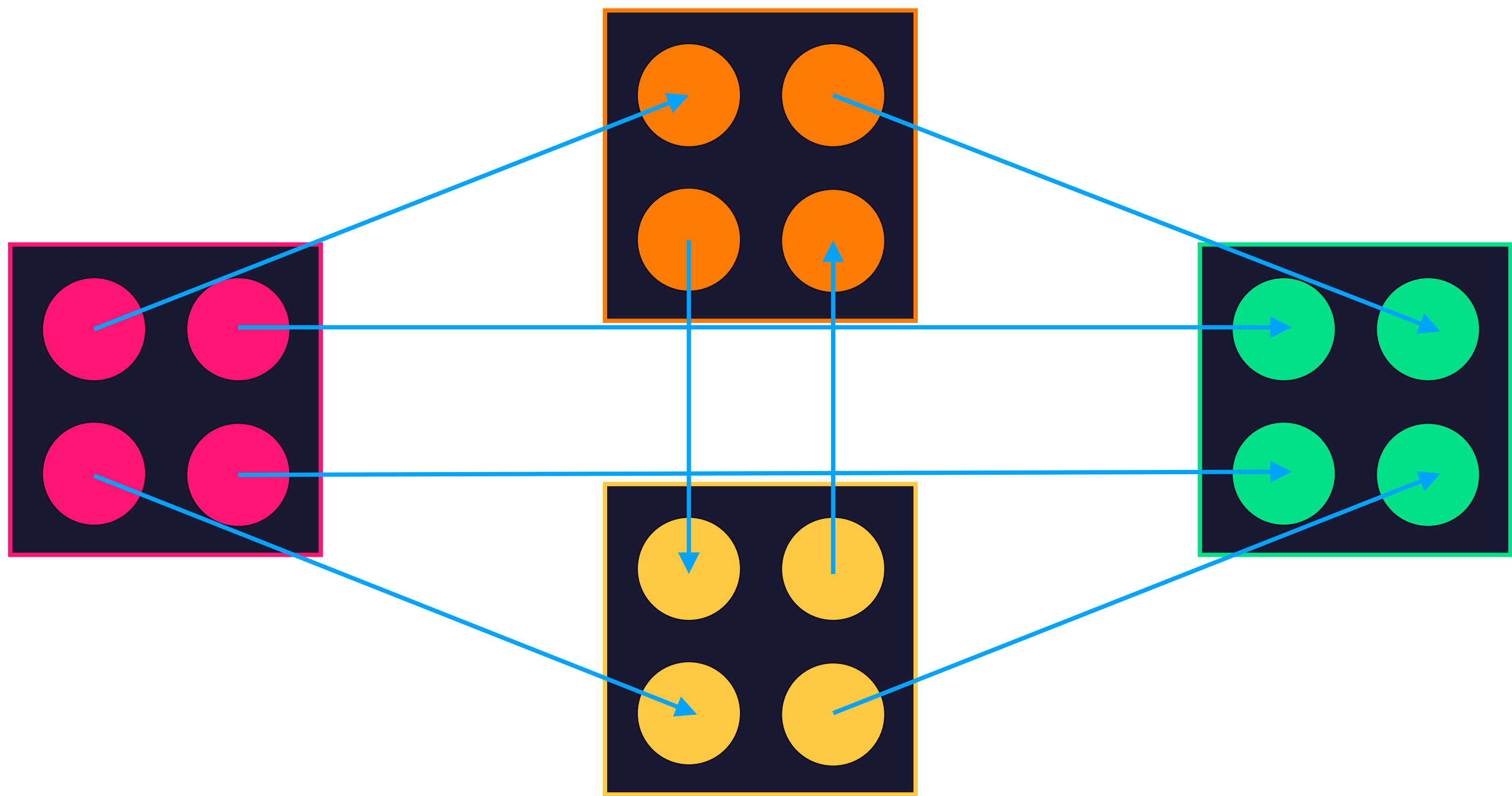




Coupling



Coupling





Nouns, Verbs, Adjectives, and Entities



Nouns and Verbs

Nouns

Classes

Verbs

Methods



Nouns = Classes



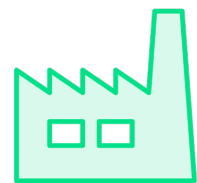
Customer – The public users of the system



Order – A heading for collections of purchased items



Shipment – Tracking details for sending orders to clients



Supplier – Manufacturer or reseller of items available to clients



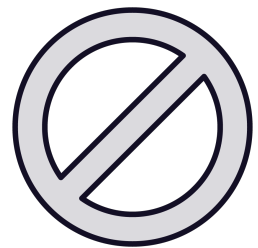
Logger – Behind-the-scenes tracing of system operations



Verbs = Methods



Ship – Send an order's items to the customer



Cancel – Stop an order from further processing



Complete – Mark an order as successfully finished



CalculateTotal – Add up the cost of an order's items and shipping



Adjectives = Interfaces



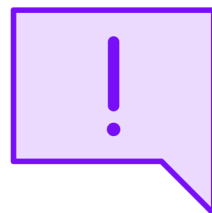
ISerializable – Object can be transported or stored



IEnumerable – Represents a collection of items that can be iterated



IDisposable – Object is responsible for cleaning up after itself



INotifyPropertyChanged – Object raises events when data changes



Parts of Speech

Nouns

Classes

Includes entities such as Customer, Order, and Product
Also includes non-entity objects like loggers, controllers, and messages

Verbs

Methods

The things your classes can *do*, both to itself or to other objects

Adjectives

Interfaces

Abstract capabilities that an object *has*, composed of its methods and properties combined



Summary



Object-oriented Design vs. Programming

Four pillars

SOLID principles

Mermaid introduction

Cohesion

Coupling

Parts of speech metaphor



Up Next:

The Four Pillars

